source ..... mutable instruments docs
rev 2026.10

plaits

_travelers guide
_for finding your voice

24	3	8
models in 3 banks	tone knobs, per-model jobs	octaves on one knob

[left] press [left] hold › then noon = 12 o'clock cw / ccw = clockwise / counter

24 models · 3 banks · each model gives the three tone knobs and aux their own jobs (atlas)

internal lpg + envelope · trig plucks · 8-octave knob · one cv per tone knob

orange = hold

01 models

plt_01

[left]	next model in the **green** bank · pitched (wraps)
[right]	next model in the **red** bank · noise + drums
[left]+[right] tap	**step mode** on / off: [left] back · [right] forward through all 24 · the only way to the **yellow** bank
LED colour	yellow = new in 1.2 · green = pitched · red = noise + drums
[model] cv	moves away from the chosen model, across all 24 · steady LED = playing · slow blink = button choice
+ trig patched	model changes wait for the next trigger
power on + [right]	colour-blind mode: brightness instead of colour

02 pitch · tone

plt_02

[frequency]	8 octaves (c0–c8) by default · range is settable (07)
[v/oct]	−3 to +7 octaves above the knob
[fm] att	how much the fm jack bends pitch (±)
[harmonics]	spread, or the balance between the parts
[timbre]	dark · sparse → bright · dense
[morph]	sideways: variations of the same colour
cv ranges	timbre fm morph ±8 v · harmo model ±5 v · level trig 0–8 v

03 trig · level · lpg

plt_03

[trig]	fires the envelope · excites physical + drum models · plucks the lpg · samples the model cv
[level]	opens the lpg (louder + brighter) · accent on physical + drum models
trig in, cv out	[timbre] [fm] [morph] atts set how much internal envelope each gets
unplugging a cv	turn its att back to noon, or the envelope takes over
red 4–8	own envelope + filter · the lpg is off · level = accent
[aux]	a variant, sidekick or by-product of out

04 green bank · pitched

plt_04

1 virtual analog — two classic waves

harmo	detune between the two
timbre	narrow pulse → square → hardsync formants
morph	triangle → saw with a widening notch
aux	two hardsynced waves
tip	narrow pulse or wide notch = silence: one osc only

2 waveshaping — triangle › shaper › folder

harmo	shaper curve
timbre	fold amount
morph	asymmetry
aux	warps-style folder

3 fm — two sines, phase-modulating

harmo	frequency ratio
timbre	modulation index
morph	feedback: ccw chaotic · cw rough
aux	sub-oscillator

4 grain formant — vosim · pulsar · cz

harmo	ratio of formant 1 to 2
timbre	formant frequency
morph	formant width + shape
aux	filtered waves · harmo = lp bp hp

5 additive — 24 harmonics

harmo	number of bumps in the spectrum
timbre	loudest harmonic (like a bp cutoff)
morph	bump shape: wide → peaked
aux	hammond drawbar harmonics only

6 wavetable — 4 banks of 8×8

harmo	bank: a b c d smooth, then d c b a stepped
timbre	row (darker → brighter)
morph	column
aux	5-bit lo-fi

7 chords — four-note string / organ

harmo	chord type
timbre	inversion + transposition
morph	string-machine waves → 16-wave table
aux	root note alone

8 speech — formant · sam · lpc

harmo	vowels, then banks of words
timbre	species: daleks → chipmunks
morph	phoneme or word
aux	raw vocal cords
words	trig speaks · fm att = intonation · morph att = speed

+ plaits palette

by rubato labs

a free browser tool from rubato labs for building your own plaits firmware. pick a model for every slot (stock models, braids models ported to plaits, alternate takes on the stock ones, experimental rubato labs models and community contributions), set the options, and generate an audio-installable build.

rubato.audio/plaits-palette · not a mutable instruments product

05 red bank · noise + drums

plt_05

1 granular cloud — swarm of 8 saws

harmo	pitch randomness
timbre	grain density
morph	grain length · cw = merged stack
aux	sine swarm
tip	supersaw: some random, some density, full overlap

2 filtered noise — clocked noise › filter

harmo	lp → bp → hp
timbre	noise clock
morph	resonance
aux	two band-passes
pitch	frequency + v/oct track the cutoff

3 particle noise — dust › resonators

harmo	frequency randomness
timbre	particle density
morph	ccw reverb-ish all-pass · cw resonant
aux	raw dust

4 inharmonic string — mini-rings red

harmo	inharmonicity / material
timbre	exciter brightness, dust density
morph	decay
aux	raw exciter
trig	none = dust excites · 3 voices

5 modal resonator — mini-rings green

harmo	material
timbre	exciter brightness, dust density
morph	decay
aux	raw exciter
voices	1 voice, 24 partials

6 bass drum — two analog flavours

harmo	attack sharpness + overdrive
timbre	brightness
morph	decay
aux	fm triangle › sine › dirty vca
trig	none = a steady tone

7 snare drum — two analog flavours

harmo	tone ↔ noise balance
timbre	balance of the drum modes
morph	decay
aux	fm sine pair + hp noise

8 hi-hat — two analog flavours

harmo	metal ↔ noise balance
timbre	high-pass cutoff
morph	decay
aux	ring-mod pairs, clean vca

red 4–8 ignore the lpg: trig strikes them, level = accent

06 yellow bank · new in 1.2

plt_06

1 va + filter — classic waves › vcf

harmo	resonance · ccw 24 dB gentle · cw 12 dB harsh
timbre	cutoff
morph	waveform + sub level
out/aux	low-pass / 12 dB high-pass

2 phase distortion — pd / pm

harmo	distortion frequency
timbre	distortion amount
morph	asymmetry
out/aux	synced carrier (pd) / free carrier (pm)

3–5 six-op fm — 2 voices · 32 presets each

harmo	preset
timbre	modulator level
morph	stretch envelopes · no trig = time-travel
trig	alternates the 2 voices · level = velocity
banks	basses + synths · keys, plucks, perc · organs, pads, strings, brass

6 wave terrain — 8 terrains, blended

harmo	terrain · fully cw = your own
timbre	path radius
morph	path offset
out/aux	height / height as phase distortion

7 string machine — stereo filter + chorus

harmo	chord
timbre	chorus / filter amount
morph	waveform
out/aux	voices 1+3 / voices 2+4

8 chiptune — four square voices

harmo	chord
timbre	arpeggio or inversion · att = env shape
morph	pulse width / sync
out/aux	squares / nes triangle
trig	clocks the arpeggiator

reached in step mode only: tap [left]+[right], then step

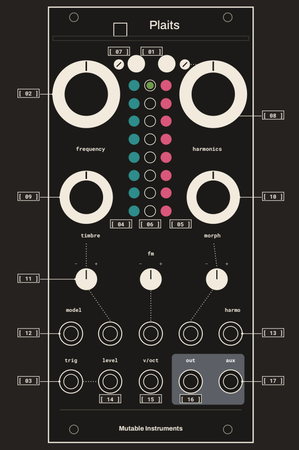
08 LED codes

plt_08

range	chasing = lfo · one LED = c0…c7 · alternating = octaves · all lit = full
lpg settings	two rows of 4 yellow: response · decay
calibration	1st LED blinks green: send 1 v · then orange: send 3 v

__fig.03 : front panel

01–07 = sections · 08–17 = hardware



[01] __buttons : models · banks
[02] __frequency : pitch · tone
[03] __trig : trig · level · lpg
[04] __teal icons : green bank
[05] __pink icons : red bank
[06] __LEDs : model · yellow bank
[07] __icons : held-button settings

[08] __harmonics : spread · balance
[09] __timbre : dark → bright
[10] __morph : sideways
[11] __atts : timbre · fm · morph
[12] __model : cv in ±5 v
[13] __cv ins : timbre fm morph harmo
[14] __level : lpg · accent
[15] __v/oct : −3 to +7 v
[16] __out : main
[17] __aux : variant

07 held buttons

plt_07

[left]+turn [timbre]	lpg response: vcfa (filter + amp) → vca
[left]+turn [morph]	lpg ring time + envelope decay
[right]+turn [harmonics]	frequency range: lfo · c0–c7 ±7 st · octaves · full 8 oct
octaves [right]+ [frequency]	set the root · then [frequency] jumps whole octaves
[left] / [right] 2 s	show the lpg / range setting
saving	settings stick only after the next model change
[left]+[right] 2 s	calibrate: 1 v › [left] › 3 v › [left]
[timbre] jack	loads custom fm patches, a terrain or waves from the online editor